

Project Objective & Data Overview

Objective:

Analyze and compare mobility patterns across NYC neighborhoods to understand their impact on local air quality. The study leverages statistical analysis, clustering, and regression techniques to identify neighborhood-specific trends and propose targeted interventions.

Data Available:

- **NYC_AQ.csv:** Air quality measurements
- **NYC_PM.csv, NYC_PM_Part1.csv, NYC_PM_Part2.csv, NYC_PM_Part3.csv:** Particulate matter (PM) and related air quality data
- **nyc_polygon.geojson:** Geospatial boundaries of NYC neighborhoods

Timeline Tasks Table

Timeline	Task	Objective	Data/Resources Used
Week 1 (Days 1–7)	- Define project objectives, scope, and research questions - Inventory available datasets - Set up project repository and tools	Establish a clear plan and understanding of the project scope; prepare the foundation for data analysis	Project documentation, meeting notes, NYC_AQ.csv, NYC_PM*.csv, nyc_polygon.geojson
Week 2 (Days 8–14)	- Import and integrate data (CSV and GeoJSON files) - Map geospatial boundaries to NYC neighborhoods - Clean and preprocess data	Ensure data is merged, cleaned, and formatted consistently; create initial features for analysis	Imported CSVs and GeoJSON; Python libraries for data cleaning and merging
Week 3 (Days 15–21)	- Perform exploratory data analysis (EDA): descriptive stats, histograms, box plots, time-series graphs - Conduct correlation analysis - Begin initial clustering/regression modeling	Identify key patterns and trends in mobility and air quality; gain insights to guide further modeling	Cleaned dataset; visualization tools (e.g., matplotlib); statistical analysis libraries
Week 4 (Days 22–30)	- Refine clustering and regression models - Develop final visualizations (bar charts, heatmaps, geospatial overlays) - Draft final report and prepare presentation	Finalize models and visuals; compile and communicate insights, trends, and neighborhood-specific recommendations	Modeling outputs; visualization tools; reporting templates and presentation software